CONCEPT PAPER · v3.0

AI-Supported Decision Assistance in Postmortem Kidney Transplantation

A Decision-Science Framework for the Center-Level Acceptance Decision in the Eurotransplant Context — with a Translational Pathway toward an MDR-Compliant Software Medical Device

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Abstract

The center-level acceptance decision in postmortem kidney transplantation is a high-stakes, time-critical clinical judgement made under incomplete information, with substantial consequences for both individual patient outcomes and population-level organ utilization. Allocation algorithms (ETKAS, ESP) within the Eurotransplant area produce a ranked list of candidate recipients but do not replace the center-level decision to accept or decline a specific organ offer. Standardized, European-context-validated decision instruments for this decision are currently lacking.

This concept paper outlines a decision-science framework for AI-supported, assistive decision assistance at the center level. The framework optimizes decision quality under uncertainty rather than the prediction of single outcomes. The primary methodological objective is the maximization of clinical net benefit per organ offer (Decision Curve Analysis); the primary clinical composite endpoint is the proportion of accepted organs with a favorable early outcome (no primary non-function, no delayed graft function, intact graft at 12 months).

A three-phase study architecture is proposed: retrospective cohort-based model development; prospective shadow-mode evaluation; and a prospective interventional pilot. A parallel translational pathway addresses MDR (Regulation (EU) 2017/745) and EU AI Act (Regulation (EU) 2024/1689) compliance, with conservative classification planning toward MDR Class IIb.

This concept paper is published as prior-art ahead of clinical implementation. It is intended to inform academic and clinical partners interested in joint multi-center evaluation.

1. Problem Statement

The acceptance decision combines heterogeneous data streams — donor parameters (age, comorbidity, cause of death, hemodynamic course, kidney function trajectory, possible histology), recipient parameters (current dialysis status, cardiovascular risk, adherence, retransplantation), the immunological constellation (HLA mismatches, indirect T-cell epitope matching, donor-specific antibodies), and logistical factors (cold ischemia time, distance, OR capacity). It must be made within a tightly bounded time window, typically under 60 minutes, often outside regular working hours.

The consequences are asymmetric and severe in both directions. A wrongful decline shortens the expected survival of the waiting patient and removes a viable organ from the system. A wrongful acceptance risks primary non-function, early graft loss, and avoidable morbidity. Schulte and colleagues, analyzing a single-center German cohort, have shown that the transferability of US-derived risk indices (KDPI, EPTS) to the Eurotransplant area is limited; among the variables encoded in KDPI/EPTS, only donor and recipient age were found to be consistently associated with graft and patient survival in their data [7]. A decision instrument specifically validated in the European context, embedded in the center-level workflow, and regulatorily robust as a software medical device does not currently exist.

2. Decision-Science Framing

The concept proposed here departs from the conventional outcome-prediction framing prevalent in the AI-in-transplantation literature. The system is explicitly designed not to predict individual outcomes but to assist the clinical decision under uncertainty. Predicted single outcomes (survival, PNF, DGF) are inputs to a decision-theoretic objective, not the objective itself.

Primary methodological objective: maximization of clinical net benefit per organ offer, evaluated against two reference strategies ("accept all offered organs" and "accept no organ") across the clinically relevant range of acceptance thresholds, using Decision Curve Analysis [1, 2].

Primary clinical composite endpoint: proportion of accepted organs with favorable early outcome, defined as the simultaneous absence of primary non-function, absence of delayed graft function, and intact graft at 12 postoperative months. This endpoint links the acceptance decision to the quality of its consequence in a clinically interpretable single measure.

Utility operationalization: the integrated utility function aggregating survival, PNF and DGF components is not specified ad hoc. Component weights are elicited in a structured Delphi panel of transplantation clinicians [3] and submitted to consistency checking via the Analytic Hierarchy Process [4]. Robustness of the resulting recommendations against weight variation is reported in a sensitivity analysis. The utility function is therefore explicit, reproducible, and auditable.

3. Three-Phase Study Architecture

3.1 Phase A — Retrospective Cohort and Model Development

Single- or multi-center retrospective cohort of postmortem kidney transplantations, typically from the past 15-20 years (anticipated cohort size 800-2,000 cases per partnering center). Methodological primacy of regularized, interpretable models — penalized Cox regression (Lasso/Elastic Net), random survival forests, gradient boosting — with an explicit regularization and hyperparameter scheme. Deeper survival architectures (e.g., DeepSurv, DeepHit) [5] are evaluated only exploratorily and in regularized form; at comparable performance, a clear precedence rule favors interpretable models. This precedence is methodologically warranted given the cohort size constraints typical at individual European transplantation centers and the regulatory importance of explainability.

Validation is performed via nested cross-validation and rolling-origin out-of-sample evaluation. Discrimination and calibration are reported per outcome component (C-index, time-dependent AUC, integrated Brier score, integrated calibration index). The primary methodological endpoint — net benefit via Decision Curve Analysis — is reported across the relevant threshold range. Explainability is provided via SHAP and Integrated Gradients with clinical plausibility cross-checks.

3.2 Phase B — Prospective Shadow Mode

The model is integrated into the clinical workflow software in a non-interventional shadow mode. For each incoming organ offer, the system generates an algorithmic recommendation and explanation, logged exclusively for study purposes and inaccessible to the clinical decision. This phase yields calibration of the model under deployment-like conditions, evaluation of recommendation stability across time and clinical subgroups, and concordance estimates between algorithmic and clinical decisions ahead of any interventional step.

3.3 Phase C — Prospective Interventional Pilot

Phase C is explicitly designed as a prospective pilot and hypothesis-generating study, not as a powered confirmatory trial. The system is made visible and usable as an assistive decision aid in the clinical workflow; the clinical team retains full decisional authority. Algorithmic recommendation and clinical decision are recorded in parallel; deviations are documented in a structured manner. Design options include single-center pre/post implementation analysis or stepped-wedge multi-center roll-out.

Primary endpoints (Phase C): the clinical composite endpoint (favorable outcome among accepted organs) and net benefit relative to the historical clinical-only baseline. Secondary process endpoints: decision concordance, change in decision behavior (acceptance/decline rates, decision time, documented confidence), inter-reader variability, and clinician acceptance. Survival endpoints are reported exploratorily; confirmatory survival evidence is reserved for a subsequent fully-powered multi-center trial.

4. Sample Size Considerations

Phase A: cohort sizes in the range of 800-2,000 transplantations per partnering center support robust internal cross-validated model development with the regularized model class proposed above. External validation against complementary center cohorts or against Eurotransplant aggregate data is foreseen as part of any multi-center extension; it is methodologically beneficial but not a precondition for the conceptual integrity of Phase A.

Phase C: assuming a baseline rate of approximately 65-70 % favorable outcome among accepted organs in historical control cohorts and a clinically relevant improvement of 10 percentage points, a one-sided test (alpha = 0.05, power = 0.80) requires roughly 200-250 transplantations during the intervention phase. This is reachable within 24-36 months at a single center with typical European case volume, or within 12-18 months in a two- to three-center consortium.

Powered survival endpoints are explicitly out of scope for Phase C and are deferred to a subsequent confirmatory trial.

5. Regulatory Framework

5.1 Medical Device Regulation

Software providing information used for diagnostic or therapeutic decisions qualifies as a software medical device under MDR [8] Article 2 and is classified under Annex VIII Rule 11 as Class IIa or higher [11]. Given the severity of consequences associated with a wrong acceptance decision (potential loss of a donor organ; potential loss of recipient life through early graft failure or waitlist mortality), the concept is positioned at the IIa/IIb boundary with conservative planning toward Class IIb. Implications include a quality management system per ISO 13485 [12], risk management per ISO 14971 [13], software life cycle per IEC 62304 [14] (Class C as planning baseline), full technical documentation, clinical evaluation per MDR Annex XIV, post-market surveillance, and a PMCF plan.

5.2 EU AI Act

An AI system that is, or is a safety component of, a medical device requiring third-party conformity assessment is classified as a high-risk AI system under the EU AI Act [9]. Resulting obligations on data governance, logging, robustness, cybersecurity, transparency, and human oversight are integrated into the existing quality management system through a unified technical documentation and a unified risk management process. The concept is also evaluated for eligibility as a use-case in an AI-Act regulatory sandbox under Article 57.

5.3 Study-Specific Regulatory Path

Phases A and B are designed as non-interventional research and do not require approval under the German Medical Devices Implementation Act (MPDG). Phase C, as an interventional study with a non-CE-marked product, is performed under §§ 40 ff. MPDG in conjunction with MDR Articles 62 ff., subject to BfArM authorization and a positive ethics committee opinion.

6. Data Governance

All personal clinical data remain under the data trusteeship of the partnering clinical institution. A pseudonymized training cohort is transferred to a secure, EU-hosted research environment for model development; the re-identification key remains exclusively at the clinical institution. No clear-text data are transferred to industry partners. Processing rests on Article 9 (2)(j) GDPR in conjunction with national research-exemption provisions, supplemented by patient information and, for prospective phases, explicit consent. Subprocessors used for model training, inference, and shadow-mode integration are restricted to EU-hosted, zero-data-retention services. The architecture is designed to be transferable to the European Health Data Space [10] secondary-use pathway as national access bodies become operational.

7. Translational Pathway

The translational pathway is structured along three parallel work streams. Phase A targets TRL 4 (laboratory validation on retrospective data); Phase B targets TRL 6 (validation in a relevant operational environment via shadow mode); Phase C targets TRL 7-8 (demonstration in the field, preparation for CE marking).

Regulatory maturation is initiated in parallel with Phase A and progresses through technical documentation v0.1 (M1-M12), clinical evaluation planning and notified-body pre-submission (M13-M24), and conformity assessment toward CE marking (M25-M36). The structural separation between an academic research entity (responsible for clinical research, publications, and non-commercial grant applications) and a commercial entity (responsible for product development, MDR conformity assessment, and commercial deployment) is maintained throughout, with arms-length licensing arrangements between the two.

8. Open Invitation to Clinical Partners

This concept is shared as the basis for academic and clinical partnership. The author seeks collaboration with one or more European transplantation centers willing to participate as clinical study sites under a structured consortium agreement. Suitable centers offer:

- An established postmortem kidney transplantation program with well-documented historical cohorts.
- Interest in evidence-based, regulatorily robust decision support — distinct from internal predictive analytics tooling.
- A clinical lead willing to assume responsibility as senior clinical investigator for the local site.
- An institutional setting that supports a non-interventional research phase followed by a regulated interventional pilot.

Detailed methodological and implementation documentation is available under a mutual non-disclosure agreement. Inquiries may be directed to the author at the contact address listed below.

9. Prior-Art Declaration

This concept paper documents the methodological framework and study architecture as authored and developed by Mohammed Alfarra in his capacity as founder of Kaun AI gGmbH and KaunSys GmbH. The deposit on Zenodo establishes a citable, time-stamped record of authorship for the concept described herein. Further iterations and version updates will be published as additional Zenodo deposits with linked DOIs. Implementation details, software architecture, and trained model artifacts are not part of this public deposit and remain proprietary intellectual property of Kaun AI gGmbH and KaunSys GmbH.

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[9] Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence and amending various other regulations (Artificial Intelligence Act).

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